1 Null Space and Column Space

Given $A \in M^{m\times n}(F)$, we have seen that we can define $T_A \in \mathcal{L}(M^{n\times 1}(F), M^{m\times 1}(F))$ by multiplying $x \in M^{n\times 1}(F)$ by the matrix A. We will now define 4 subspaces, 2 of $M^{n\times 1}(F)$ and 2 of $M^{m\times 1}(F)$, associated to A called the Fundamental Matrix Spaces.



Definition 1.1 (Null space of A, Nullity of A)

The *null space of* A, denoted by Nul$A$, is the kernel of the linear map T_A. That is,

$$\begin{aligned}\text{Nul}A := \ker T_A &= \{x \in M^{n\times 1}(F) : T_A x = 0\} \\ &= \{x \in M^{n\times 1}(F) : Ax = 0\},\end{aligned}$$

which is the solution set of the homogenous linear system $Ax = 0$. The dimension of Nul$A$ is called the *nullity of* A.



Definition 1.2 (Column space of A)

We define the *column space of* A, denoted by Col$A$, to be the image of the linear map T_A. Denoting the i-th column of the matrix A by $A_{\cdot,j}$, we have:

$$\begin{aligned}\text{Col}A := T_A &= \{T_a x : x \in M^{n\times 1}(F)\} \\ &= \{Ax \in M^{n\times 1}(F)\} \\ &= \{\sum_{i=1}^{n} x_i A_{\cdot,i} : x_i \in F\} \\ &= \qquad\qquad\qquad\qquad \{a_{\cdot,1,\ldots,A_{\cdot,n}}\end{aligned}$$

that is, the column space of A is the span of the column vectors of the matrix A. The dimension of the column space of A is called the *rank of* A.

By the Fundamental Theorem of Linear Maps:

$$\dim M^{n\times 1}(F) = \dim \ker T_A + \dim T_A \tag{1}$$
$$n = \dim \text{Nul}A + \dim \text{Col}A \tag{2}$$
$$n = \text{nullity of } A + \text{ rank of } A. \tag{3}$$

Suppose first that A is in RREF.

- Since each non-pivot column is a linear combination of the pivot columns to the left of it, by the plus/minus lemma, the non-pivot columns and be removed from the set of column vectors without changing Col$A$. Since the set of pivot columns is linearly independent (by definition of RREF), this implies that rank$A$ = number of pivot columns.
- Each non-pviot column gives rise to a free variable. As we have seen, the dimension of the solution set of the system $Ax = 0$ is equal to the number of free variables, so nullity A = # of non-pivot columns. Thus, the equation $n = \text{nullity}A + \text{rank}A$ simply expresses the fact that the total number of columns of A is the sum of the pivot and non-pivot columns.

If A is not in RREF, then it is still true that the pivot columns of A are a basis of Col$A$. This is because elementary row operations do not change the linear dependence relations among the columns of A, since if two matrices are row-equivalent, $Ax = 0$ and $Bx = 0$ have the same solution sets.

These observations give rise to an algorithm for producing explicitly bases of Nul$A$ and Col$A$:

1. Row-reduce A to its RREF, R
2. Use R to identify the pivot columns of A, which are a basis of Col$A$.
3. Use R to find a parametric description of the solution set of $Ax = 0$. The vectors spanning the solution set are a basis of Nul$A$.

Example 1.3

Consider the matrix

$$A = \begin{bmatrix} -3 & 7 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

The reduced row echelon form of A is

$$R = \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The first and third columns of A are pivot columns, so a basis of Col$A$ is given by:

$$\{ \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}.$$

Since x_2, x_4, x_5 are free variables, we set

$$\begin{aligned} x_2 &= t_1 \\ x_4 &= t_2 \\ x_5 &= t_3. \end{aligned}$$

A vector $x \in F^5$ is therefore a solution to this system of equations if it is of the form

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2t_1 + t_2 - 3t_3 \\ t_1 \\ -2t_2 + 2t_3 + t_2 + t_3 \\ t_2 \\ t_3 \end{bmatrix} = t_1 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t_3 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}.$$

therefore a basis of Nul$A$ is given by

$$\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \}$$

Exercise 1.4

Find explicit bases of the null space and column space of the matrix

$$A = \begin{bmatrix} 1 & 4 & 0 & 2 & -1 \\ 3 & 12 & 1 & 5 & 5 \\ 2 & 8 & 1 & 3 & 2 \\ 5 & 20 & 2 & 8 & 8 \end{bmatrix}$$



Solution

Row-reducing A, we get:

$$R = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

So, the 2nd and 4th columns are free variables, and so the solution set to $Rx = 0$ is given by:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = t_2 \begin{bmatrix} -4 & 1 & 0 & 0 & 0 \end{bmatrix} + t_1 \begin{bmatrix} -2 & 0 & 1 & 0 & 0 \end{bmatrix},$$

so the nullspace is spanned by:

$$\{\begin{bmatrix} -4 & 1 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} -2 & 0 & 1 & 0 & 0 \end{bmatrix}\}$$

The column space is spanned by the pivot columsn in the original matrix, or:

$$\{\begin{bmatrix} 1 \\ 3 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 5 \\ 2 \\ 8 \end{bmatrix}\}.$$

2 Row Space and Left Null Space



Definition 2.1 (Transpose of a matrix)

If A is a $m \times n$ matrix, then A^T denotes the *transpose* of A, which is the $n \times m$ matrix obtained by interchanging rows and columns of A.



Example 2.2

If $A = \begin{bmatrix} 2 & 1 & -3 \\ 5 & 6 & 7 \end{bmatrix}$, then $A^T = \begin{bmatrix} 2 & 5 \\ 1 & 6 \\ -3 & 7 \end{bmatrix}$.



Definition 2.3 (Row space of A)

We define the *row space of* A to be the span of the row vectors of the matrix A. Denoting the ith row of A by $A_{i,\cdot}$:

$$\text{Row}A := \{A_{1,\cdot}, ..., A_{m,\cdot}\}.$$

Note that Row A = Col A^T.

Suppose A is in RREF.

- By definition of RREF, the first row is nonzero, and the remaining nonzero rows are not linear combination of the previous ones. Thus, these are linearly independent and therefore are a basis of Row A.
- Since each nonzero row contains a pivot, this shows that $\dim \text{Row}A = \#$ of pivots $= \dim \text{Col}A = \text{rank}A$.

Suppose now that A is not in RREf. Let R be the RREF of A.

- Since elementary row operations are invertible, they don't change the row space, so we conclude that the nonzero rows of R are a basis of Row$A$.
- Since elementary row operations don't change pivot positions, we againc conclude that $\dim \text{Row}A = \dim \text{Col}A = \text{rank}A$.



Example 2.4

Recall from Example 1.3 the matrix:

$$A = \begin{bmatrix} -3 & 7 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

which has row echelon form

$$R = \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

A basis for Row A is given by the nonzero rows of R:

$$\{ \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \\ -2 \end{bmatrix} \}$$



Definition 2.5 (Left null space)

The *left null space of* A is defined as the null space of A^T. That is, $\text{Nul}(A^T)$ is the solution set of the homogenous system $A^T y = 0$, or equivalently, the system $y^T A = 0^T$ (which is why it is called the left null space).

We have just seen that we can find a basis of $\text{Row}A = \text{Col}A^T$ without row-reducing A^T. We can also find a basis of $\text{Nul}A^T$ without having to also row-reduce A^T. We can view $y^T A$ as a linear combination of the rows of A:

$$y^T A = \sum_{i=1}^{n} y_i A_{i,\cdot}$$

The condition that y is in the left null space of A is that this linear combination is the zero (row) vector. Suppose we form the augmented matrix $[A \mid b]$ before row-reducing to RREf. There will be $n - \text{rank}A$ zero rows, which is exactly the dimension of the left null space. Each zero row of A will be of the form $[0, 0, ..., 0 \mid \alpha_1 b_1 + ... + \alpha_m b_m]$ for some scalars $\alpha_1, ..., \alpha_m$. For each zero row, we have found a linear combination of the rows of A (by the sequence of elementary row operations reducing A to RREF) which has produced the zero row. The vector of coefficients $[\alpha_1, ..., \alpha_m]$ is therefore an element of the left null space of A. It follows from the definition of RREF that these vectors are linearly independent and hence a basis of the left null space of A.



Example 2.6

We will find explicit bases for all 4 fundamental matrix spaces of the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 2 & 4 \\ 2 & 5 \end{bmatrix}$. Add an augmented column and row-reduce to RREF:

$$A = \begin{bmatrix} 1 & 2 & b_1 \\ 2 & 3 & b_2 \\ 2 & 4 & b_3 \\ 2 & 5 & b_4 \end{bmatrix} \rightarrow R = \begin{bmatrix} 1 & 0 & 5b_1 - 2b_4 \\ 0 & 1 & -2b_1 + b_4 \\ 0 & 0 & -2b_1 + b_3 \\ 0 & 0 & -4b_1 + b_2 + b_4 \end{bmatrix}.$$

A is a 4 by 2 matrix. The rank of A is 2, so the nullity is zero by rank-nullity.
A^T is a 2×4 matrix. Since rank A^T = rank $A = 2$, the left null space has dimension 2 by rank-nullity.

- A basis of Col A is given by the pivot columsn of A, $\{\begin{bmatrix}1 & 2 & 2 & 2\end{bmatrix}, \begin{bmatrix}2 & 3 & 4 & 5\end{bmatrix}\}$
- Nul $A = \{\begin{bmatrix}0 & 0\end{bmatrix}\}$, so a basis is given by $\{\}$.
- A basis of Row A is given by the nonzero rows of R: $\{\begin{bmatrix}1 & 0\end{bmatrix}, \begin{bmatrix}0 & 1\end{bmatrix}\}$.
- The entries in the augmented columns of the zero rows of R are $-2b_1 + b_3$ and $-4b_1 + b_2 + b_4$. The coefficient vectors give a basis of the left null space of A: $\{\begin{bmatrix}-2 & 0 & 1 & 0\end{bmatrix}, \begin{bmatrix}-4 & 1 & 0 & 1\end{bmatrix}\}$.



Exercise 2.7

Find explicit bases of all four fundamental matrix spaces of the matrix:

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}.$$



Solution

This row-reduces to:

$$\begin{bmatrix} 1 & -2 & 0 & 0 & -2 & \frac{1}{2}(-x - 5y + 2z) \\ 0 & 0 & 1 & 0 & 8 & x + 13y - 5z \\ 0 & 0 & 0 & 1 & -5 & \frac{1}{2}(-x - 15y + 6z) \end{bmatrix}$$

. From this, we see:

$$\text{Col}A = \{ \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 8 \end{bmatrix} \}.$$

We have two free variables, and we see the solution set to $Rx = 0$ is of the form:

$$\{ \begin{bmatrix} x_1 \\ x_2 \\ X_3 \\ x_4 \\ x_5 \end{bmatrix} = t_1 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} 2 \\ 0 \\ 0 \\ -8 \\ -5 \end{bmatrix} \},$$

so we have

$$\text{Nul}A = \{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \\ -8 \\ -5 \end{bmatrix} \}\}$$

and we get:

$$\text{Row}A = \{\begin{bmatrix}1 & -2 & 0 & 0 & -2\end{bmatrix}, \begin{bmatrix}0 & 0 & 1 & 0 & 8\end{bmatrix}, \begin{bmatrix}2 & -4 & 5 & 8 & -4\end{bmatrix}$$

and:

$$\text{Left Null}A = \{\begin{bmatrix}-\frac{1}{2} & -5 & 2\end{bmatrix}, \begin{bmatrix}1 & 13 & -5\end{bmatrix}, \begin{bmatrix}-\frac{1}{2} & -\frac{15}{2} & 3\end{bmatrix}\}.$$